Total marks: 24

Approximate guide to what the marks stand for:

M= a mark associated with applying an appropriate method

J= a mark associated with using correct justification

A= a mark associated with a correct answer or intermediate step

MAST20026 Real Analysis Semester 2 2025

Assignment 1

Due: 14 August, 11:59pm

Please see the assignment instructions on the Assignment 1 page on the LMS, where you accessed this file. To ease submission, please prepare assignment solutions for each question on its own page. This document is 2 pages in length.

PART A (*2 marks* total)

The goal of assessments in this course is to help you know how well you are mastering the concepts presented in lecture and tutorials. The work that you hand in for assessments is meant to reflect your level of understanding. And so it is important that the work you hand in fully demonstrates mastery of subject concepts.

Download the filed called *Guidelines for Good Mathematical Writing* from the Assignment 1 page on the LMS. Read it and write a short reflective response (around half a page) that considers the following questions.

- Was there anything about the reading that you found surprising?
- What experiences, if any, have you had in mathematics classes where you were expected to write in full sentences as part of your solutions? Do you find it difficult to write in mathematics courses?
- How will your approach to crafting your submission for this assignment change, if at all, as a result of having done this reading?

1 mark for demonstrating thoughtful engagement with the task.

1 mark for clearly communicated response (the response is well written, and conveys thoughts concisely).

PART B (*23 marks* total)

1. **Converse**. Let p, q be statements and consider the compound statement:

$$(p \implies q) \implies ((\sim p) \implies (\sim q))$$

   (a) (*2 marks*) Use a truth table to determine whether $(p \implies q) \implies ((\sim p) \implies (\sim q))$ is a tautology, contradiction, or neither? Briefly explain your reasoning by referring to specific lines of your truth table from Part (a).

   (b) (*2 marks*) If p is false and $(p \implies q) \implies ((\sim p) \implies (\sim q))$ is true, what can you deduce about the truth of q? Briefly explain your reasoning by referring to specific lines of your truth table from Part (a).

   (c) (*3 marks*) Write down the converse for $(p \implies q) \implies ((\sim p) \implies (\sim q))$. Furthermore, write down another logically equivalent statement to the converse, using as few

connectives $\sim, \wedge, \vee, \implies, \iff$ as possible. You can use as many parentheses as needed. You should justify why the number of connectives that you've used is minimal.

Solution.

(a) This is a truth table:

p	q	$\sim p$	$\sim q$	$p \implies q$	$(\sim p) \implies (\sim q)$	$(p \implies q) \implies ((\sim p) \implies (\sim q))$
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

1A: Correct truth table

The statement is neither a tautology nor a contradiction.
Since $(p \implies q) \implies ((\sim p) \implies (\sim q))$ is false in the third line of the truth table but true in the rest, $(p \implies q) \implies ((\sim p) \implies (\sim q))$ is neither a tautology nor contradiction. 1J: Not a tautology, nor a contradiction, with valid reason.

(b) The statement q must be false. 1A: q must be False
The only line on the truth table above where p is false and $(p \implies q) \implies ((\sim p) \implies (\sim q))$ is true is the fourth line.
On this line, q is false. 1J: Valid reason that references the truth table

(c) The converse is: $((\sim p) \implies (\sim q)) \implies (p \implies q)$.1A: Correctly written converse
We draw a truth table for the converse:

p	q	$\sim p$	$\sim q$	$p \implies q$	$(\sim p) \implies (\sim q)$	$((\sim p) \implies (\sim q)) \implies (p \implies q)$
T	T	F	F	T	T	T
T	F	F	T	F	T	F
F	T	T	F	T	F	T
F	F	T	T	T	T	T

We note that the converse is not logically equivalent to p or q since they do not have the same truth values in the truth table.
However, by comparing with $p \implies q$, we see that it *does* have the same truth values as the converse so: 1M: Valid method for determining minimal number of connectives

$$[((\sim p) \implies (\sim q)) \implies (p \implies q)] \equiv p \implies q.$$

We have used 1 connective in writing $p \implies q$ and explained why we cannot just use 0 connectives since the converse is not logically equivalent to p, and is not logically equivalent to q. Hence 1 connective is minimal. 1J: One connective needed, with complete valid reason

2. **Equivalence relation ideas.** Consider the condition:

$$p(x, y) : x - y \in \mathbb{Z}$$

(a) (*2 marks*) Translate the statement $(\forall x \in \mathbb{Z})\ [(\exists y \in \mathbb{Z})\ p(x, y)]$ into an English sentence.

(b) (*2 marks*) Is the statement $(\forall x \in \mathbb{R})\ [(\exists y \in \mathbb{R})\ p(x, y)]$ true or false? Briefly explain.

(c) (*2 marks*) Is the statement $(\exists y \in \mathbb{R})\ \ [(\forall x \in \mathbb{R})\ \ p(x,y)]$ true or false? Briefly explain.

(d) (*3 marks*) Translate the statement below into and English sentence and provide an (informal) proof of it:

$$(\forall x, y, z \in \mathbb{R})\ [p(x,y) \wedge p(y,z) \implies p(x,z)]$$

Solution.

1A: Correct or mostly correct statement in (mostly) words.
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(a) For every integer x there exists an integer y such that $x - y$ is an integer.

(b) The statement is True. 1A: The statement is true
Here's a brief argument:
Let $x \in \mathbb{R}$.
Let $y = x$.
Then $p(x,y)$ is the statement that $0 \in \mathbb{Z}$.
Since 0 is an integer, $p(x,y)$ is true. 1J: A valid reason / argument.

(c) The statement is False. 1A: The statement is false
Here's a brief argument:
Let $y \in \mathbb{R}$.
To show the statement is false, we just need a counter-example to $(\forall x \in \mathbb{R})\ \ p(x,y)$.
Let $x = y + \pi$.
Then $p(x,y)$ is the statement that $y - y + \pi \in \mathbb{Z}$, which is false since $\pi \notin \mathbb{Z}$.
So $(\exists y \in \mathbb{R})\ \ [(\forall x \in \mathbb{R})\ \ p(x,y)]$ is false. 1J: A valid reason / argument.

(d) An english sentence could be:
For every triple of real numbers x, y and z, if $x - y$ and $y - z$ are integers, then $x - z$ is also an integer. 1A: A correct statement in (mostly) words.

Here's an informal proof:

Let $x, y, z \in \mathbb{R}$. 1M: Valid approach for the proof.
Assume $x - y \in \mathbb{Z}$ and $y - z \in \mathbb{Z}$.
Then $x - z = (x - y) + (y - z) \in \mathbb{Z}$ (since $\mathbb{Z}$ is closed under addition)
Therefore $x - z \in \mathbb{Z}$. 1J: A correct proof with an indication as to why the sum is an integer.

3. **Perfect Squares.** Consider the statement:

The difference of the squares of two odd integers is even.

(a) (*2 marks*) Translate the above statement into the language of formal logic and mathematics.

(b) (*5 marks*) Write a (semi-)formal proof of the above statement.

Solution.

(a)

$$(\forall x, y \in \mathbb{Z})\ \ [[[(\exists j \in \mathbb{Z})\ \ x = 2j + 1] \wedge [(\exists k \in \mathbb{Z})\ \ y = 2k + 1]] \implies [(\exists \ell \in \mathbb{Z})\ \ x^2 - y^2 = 2\ell]].$$

1A: A correct or mostly correct statement in formal logic and mathematics that translates the English statement.

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Remarks: Other ways of writing this are possible. Note that the word difference could also reasonable be interpreted as $|x^2 - y^2|$ rather than $x^2 - y^2$.

(b) *Proof.*

1.	Let x and y be odd integers.	(premise)
2.	$(\exists j \in \mathbb{Z})\ \ x = 2j + 1$	(1., defn of odd)
3.	$(\exists k \in \mathbb{Z})\ \ y = 2k + 1$	(1., defn of odd)
4.	$x^2 - y^2 = (2j + 1)^2 - (2k + 1)^2$	(2., 3.)
5.	$x^2 - y^2 = 2(2j^2 + 2j - 2k^2 - 2k)$	(4., algebra)
6.	$(\exists \ell \in \mathbb{Z})\ \ x^2 - y^2 = 2\ell$	(5., $\mathbb{Z}$ closed under addition)
7.	$x^2 - y^2$ is even.	(6., defn of even)
8.	$(\forall x, y \in \mathbb{Z})\ ((x \text{ is odd}) \vee (y \text{ is odd})) \implies (x^2 - y^2) \text{ is even}$	(1., 7.)

Remarks: Other proofs are possible. The justification in line 6. could reasonably be considered algebra, noting that $l = 2j^2 + 2j - 2k^2 - 2k \in \mathbb{Z}$.

1M: A demonstrated understanding of what a (semi-)formal proof is.

1M: Demonstrated understanding of how a correct proof of the statement should proceed (eg. a “scratch proof” with correct ideas
could demonstrate this. A correct proof also demonstrates this)

1J: The statements / conditions in the lines of the proof follow a logical order and are mostly correct. (At most 2 minor errors)

1J: The justifications for each line in the proof mostly reference the correct previous lines, and definitions, and explain the current line. (At most 2 minor errors)

1J: A complete and correct (semi-)formal proof, with work written out clearly.